Siddhartha Shankar Das

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INTERESTS

I am a Post-Doctorate Researcher at Pacific Northwest National Laboratory (PNNL) in the **Language Intelligence** group within the **Data Science and Machine Intelligence** department. My research focuses on artificial intelligence and machine learning for scientific and applied domains, with interests spanning graph neural networks, graph sparsification, natural language processing, cybersecurity, and scalable AI systems. I develop learning-based methods for system design and optimization, including hardware/software partitioning and efficient learning on large graphs. I also build LLM-based, multimodal, and agentic AI systems for document intelligence, retrieval-augmented generation, georeferencing, and interactive decision support. My work has been published in top-tier venues and has received best paper awards.

EDUCATION

Purdue University, West Lafayette, IN, USA
Ph.D. in Computer Science — Aug 2018 – May 2025
Advisor: Alex Pothen
Thesis: Scaling Graph Neural Networks for Large Graphs and Analyzing Common Vulnerabilities via Large Language Models [pdf].
Topics: Machine Learning on Graphs, Graph Neural Networks, Cybersecurity, Large Language Models

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
M.Sc. in Computer Science and Engineering — Sep 2017
GPA: 4.00/4.00
Thesis: Evolutionary Algorithm Using Adaptive Fuzzy Dominance and Reference Point for Many-Objective Optimization (F-DEA) [pdf].
Advisor: Md. Monirul Islam
Topics: Multi-objective Optimization, Evolutionary Algorithms

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering — Jul 2014
GPA: 3.88/4.00, Major: 3.99/4.00
Thesis: A Many-objective Evolutionary Approach Using Fuzzy Dominance with Bidirectional Bias [pdf].
Advisor: Md. Monirul Islam
Topics: Multi-objective Optimization, Evolutionary Algorithms

WORK EXPERIENCE

Postdoctoral Researcher, Data Science & Machine Intelligence — May 2025 – Present
Pacific Northwest National Laboratory (PNNL), Richland, WA, USA.

- Conduct research in artificial intelligence, machine learning, graph algorithms, and system design, with a focus on scalable Graph Neural Network (GNN)-based methods for optimization problems such as hardware/software partitioning, scheduling, graph partitioning, and graph sparsification.
- Develop LLM-based, multimodal, and agentic AI systems for scientific and applied applications, including large-scale processing of environmental and regulatory documents. Current work includes

metadata extraction, georeferencing, project summarization, retrieval-augmented generation, and interactive decision-support systems.

- Contribute to interdisciplinary projects spanning cybersecurity, document intelligence, and scientific machine learning, translating foundational methods into scalable tools, datasets, deployable systems, and research publications.

Graduate Research Assistant Aug 2018 – May 2025
Department of Computer Science, Purdue University, West Lafayette, IN, USA.

- Conducted research on scalable graph learning, graph sparsification, and machine learning on large graphs, along with LLM-based methods for software vulnerability analysis and cyber threat understanding.
- Developed new sampling and sparsification methods for Graph Neural Networks, leading to publications in venues including KDD, DSAA, Big Data, and HST.

Ph.D. Intern, Data Analytics Summers 2019–2021
Pacific Northwest National Laboratory (PNNL), Richland, WA, USA
Mentored by Mahantesh Halappanavar.

- Developed learning-based methods for classification and analysis of software vulnerabilities, including mapping CVEs to weaknesses (CWEs) and attack patterns (CAPECs).
- Used Large Language Models (LLMs) and Natural Language Processing (NLP) techniques, and scaled these methods for modern AI accelerators.
- Published four papers during the internship period, including work recognized with two best paper awards.

Assistant Professor, Department of Computer Science and Engineering Dec 2017 – Jul 2018
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Roles: University teaching, project supervision, and office administration.
Study Leave for Ph.D. Jul 2018 – Jul 2024

Lecturer, Department of Computer Science and Engineering Aug 2014 – Dec 2017
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
Roles: University teaching.

SELECTED PUBLICATIONS (FULL LIST AT GOOGLE SCHOLAR)

Conference and Workshop Publications

- **Siddhartha Shankar Das**, N. Arafat, M. Rahman, S. M. Ferdous, A. Pothen, M. Halappanavar, and D. Rawat. "SGS-GNN: A Supervised Graph Sparsifier for Graph Neural Networks." *In Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD 2026), August 9–13, 2026, 12 pages.* [doi]
- **Siddhartha Shankar Das**, Zakaria Mehrab, James Kotary, Rounak Meyur, S. M. Ferdous, Erdal Mutlu, and Mahantesh Halappanavar. "Diff-GNN: Differentiable Graph Neural Networks for Joint Hardware–Software Partitioning and Scheduling." Accepted at HPEC 2026.
- James Kotary, Zakaria Mehrab, **Siddhartha Shankar Das**, Rounak Meyur, S. M. Ferdous, Erdal Mutlu, and Mahantesh Halappanavar. "Joint Optimization of HW/SW Partitioning and Scheduling with Deep Reinforcement Learning." Accepted at HPEC 2026.

- Moqsadur Rahman, Monika Akbar, Joseph T. Aguayo, Mahmud Shahriar Hossain, Tina M. Ellis, **Siddhartha Shankar Das** et al. "ATT&CK to CVE: A Large-Scale Automated Knowledge Graph for Threat Intelligence." IEEE International Conference on Data Mining (ICDM) GTA'3 Workshop, 2025. [paper]
- **Siddhartha Shankar Das**, S. M. Ferdous, M. Halappanavar, E. Serra, and A. Pothen. "AGS-GNN: Attribute-guided Sampling for Graph Neural Networks." In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024. [paper]
- **Siddhartha Shankar Das**, E. Serra, M. Halappanavar, A. Pothen, and E. Al-Shaer. "V2W-BERT: A Framework for Effective Hierarchical Multiclass Classification of Software Vulnerabilities." In *2021 IEEE 8th International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 1–12, 2021. [doi] [**Best Application Paper**]
- **Siddhartha Shankar Das**, A. Dutta, S. Purohit, E. Serra, M. Halappanavar, and A. Pothen. "Towards Automatic Mapping of Vulnerabilities to Attack Patterns Using Large Language Models." In *2022 IEEE International Symposium on Technologies for Homeland Security (HST)*, 2022. [doi] [**Best Paper**]
- **Siddhartha Shankar Das**, M. Halappanavar, A. Tumeo, E. Serra, A. Pothen, and E. Al-Shaer. "VWC-BERT: Scaling Vulnerability–Weakness–Exploit Mapping on Modern AI Accelerators." In *2022 IEEE International Conference on Big Data*, pp. 1224–1229, 2022. [doi]
- K. Panchal, **Siddhartha Shankar Das**, L. Torre, J. Miller, R. Rallo, and M. Halappanavar. "Efficient Clustering of Software Vulnerabilities Using Self-Organizing Maps (SOM)." In *2022 IEEE International Symposium on Technologies for Homeland Security (HST)*, pp. 1–7, 2022. [doi]
- A. Rizvi, T. R. Toha, **Siddhartha Shankar Das**, Sriram Chellappan, A. B. M. Alim Al Islam. "Many-objective Performance Enhancement in Computing Clusters." In *2017 IEEE 36th International Performance Computing and Communications Conference (IPCCC)*, 2017. [doi]
- A. Rizvi, T. R. Toha, **Siddhartha Shankar Das**, Sriram Chellappan, A. B. M. Alim Al Islam. "Exploiting a Synergy between Greedy Approach and NSGA for Scheduling in Computing Clusters." In *2018 5th International Conference on Networking, Systems and Security*, 2018. [doi]

Journal Articles, Book Chapters, and Reports

- Moqsadur Rahman, Aaron Sanchez, Krish Piryani, **Siddhartha Shankar Das**, Sai Munikoti, Luis de la Torre Quintana, Monowar Hasan, Joseph Aguayo, Monika Akbar, Shahriar Hossain, and Mahantesh Halappanavar. "Generative AI for Advanced Cyber Defense." *AI for Cybersecurity: Research and Practice*, John Wiley & Sons, Inc., pp. 109–146, 2025. [book chapter]
- Sai Munikoti, Dan Nally, Sai Dileep Koneru, **Siddhartha Shankar Das**, et al. "NEPATEC v2.0: Standardized Metadata and Text Corpus of National Environmental Policy Act Documents." Pacific Northwest National Laboratory, 2025. [report]
- **Siddhartha Shankar Das**, Md. Monirul Islam, Naheed Anjum Arafat. "Evolutionary Algorithm Using Adaptive Fuzzy Dominance and Reference Point for Many-Objective Optimization." *Swarm and Evolutionary Computation*, 44:1092–1107, 2019. [doi]

Preprints

- **Siddhartha Shankar Das**, S. K. Navuluru, S. M. Ferdous, R. A. Rossi, B. Coskunuzer, L. Tamil, E. Serra, A. Pothen, R. Rallo, and M. M. Halappanavar. "Scaffold: Support Graph Theory Based Sparsification for Graph Neural Networks." *arXiv:2609.31466*, 2026. [arxiv]
- S. K. Navuluru, **Siddhartha Shankar Das**, F. Dernoncourt, S. M. Ferdous, R. A. Rossi, N. K. Ahmed, B. Coskunuzer, A. Pothen, L. Tamil, and M. M. Halappanavar. "EDiS: Edge Disjoint

Subgraph Sparsification Framework for Graph Neural Networks." *arXiv:2610.09059*, 2026. [arxiv]

- S. K. Navuluru, **Siddhartha Shankar Das**, B. Ni, H. Chen, Y. Wang, B. Coskunuzer, N. K. Ahmed, F. Dernoncourt, M. Halappanavar, T. Derr, R. A. Rossi, and L. Tamil. "Unifying Graph Neural Networks Through a Common Layer Equation." *arXiv:2608.16097*, 2026. [arxiv]

ONGOING AND PAST PROJECTS

Graph Learning, System Design, and Optimization

- **Graph-Based Circuit Equivalence Checking:** Developing graph learning methods for circuit equivalence verification using GNN-based subcircuit identification, gate embeddings, and graph matching.
- **Benchmarking Graph Sparsification Methods and Applications:** Developing a comprehensive textbook and benchmarking framework covering graph sparsification, rewiring, augmentation, theoretical guarantees, and applications.
- **Graph Neural Network Architecture Generation:** Exploring automated GNN architecture generation through a unified framework for message passing, aggregation, and transformation.

NEPA, Multimodal AI, and Agentic Systems

- **NEPATEC 2.0: Standardized NEPA Document Corpus:** Contributed to a publicly available NEPA dataset for document intelligence, released on Hugging Face.
- **NEPATEC 2.5: Multimodal Metadata Extraction:** Developing multimodal LLM methods for NEPA metadata extraction, map-based geolocation, and project summarization.
- **PermitCE: AI-Assisted Chat for NEPA Documents:** Developing a retrieval-augmented generation system for semantic search, recommendations, summarization, and question answering.

Cybersecurity and Applied Machine Learning

- **Composite Metric for Cyber Risk Prioritization:** Developing a vulnerability prioritization metric combining CVSS, CWSS, EPSS, and LEV.
- **Predictive Analytics for Russet Potato Selection:** Applying machine learning and feature selection to phenotypic data for potato breeding decisions.
- **Machine Learning for USEPA SPECIATE Database:** Applying clustering methods to identify similarities among chemical source profiles.
- **Miscellaneous Collaborative Research:** Collaborating on interdisciplinary research in graph learning, cybersecurity, environmental science, and applied AI.

More Projects: A complete list of my other projects is available here.

TEACHING AND MENTORING

Pacific Northwest National Laboratory (PNNL) Aug 2025 – May 2026
Research Mentoring: Mentored five interns over the past year on graph neural networks, circuit representation learning, and cybersecurity, resulting in three accepted publications.

Purdue University, West Lafayette, IN, USA Aug 2018 – May 2019
Department of Computer Science
Position: Graduate Research Assistant.
Courses: Algorithms, Data Structures, etc.

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh Aug 2014 – Jul 2018
Department of Computer Science and Engineering
Position: Assistant Professor (2017-2018), Lecturer (2014-2017).
Courses: Machine Learning, Artificial Intelligence, Computer Graphics, Data Communication, and others (List of Courses)

SKILLS

Research Areas:	AI/ML, Graph neural networks, Graph Algorithms, Graph Sparsification, Natural Language Processing, Cybersecurity, Scientific Machine Learning.
Languages:	Python, C/C++, Java, MATLAB, Julia, C#, SQL, JavaScript.
Frameworks/Tools:	PyTorch, PyTorch Geometric, DGL, TensorFlow, Keras, MPI, Git, Linux, MOEA Framework.
LLM and AI Systems:	Large language models, multimodal AI, agentic AI, retrieval-augmented generation (RAG), document intelligence, metadata extraction, georeferencing, interactive AI assistants.
Systems and Applications:	System design, hardware/software partitioning, scalable AI systems, large-scale document processing, vulnerability analysis, CVE–CWE–CAPEC mapping, cyber risk prioritization.
Databases:	Oracle, Microsoft SQL Server, MySQL, SQLite.

ACADEMIC SERVICES

Reviewer: ICML'25, KDD'24, DSAA'23, The Web Conference (formerly WWW), IEEE Transactions on Artificial Intelligence and other venues.

Organizing Member: NSysS 2015, 2016, 2017.

AWARDS AND ACHIEVEMENTS

Best Application Paper Award, DSAA 2021
Best Paper Award, HST 2022
Dean's Honor List, BUET, 2011–2014
University Merit Scholarship, BUET

REFERENCES

Alex Pothen
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